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(12) **United States Design Patent** (10) **Patent No.:** **US D1,088,267 S**
Chang et al. (45) **Date of Patent:** **** Aug. 12, 2025**

(54) **BIOLOGICAL CELL PROCESSING CHIP**

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(**) Term: **15 Years**

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Related U.S. Application Data

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(51) **LOC (15) Cl.** **24-02**

(52) **U.S. Cl.**
USPC **D24/225**

(58) **Field of Classification Search**
USPC D24/168, 169, 186, 187, 200, 216,
D24/223–227, 230, 232, 233; D10/81
(Continued)

(56) **References Cited**

U.S. PATENT DOCUMENTS

5,955,028 A 9/1999 Chow
6,042,709 A * 3/2000 Parce B01L 3/502715
204/600

(Continued)

FOREIGN PATENT DOCUMENTS

CN 101928666 B 4/2013
CN 106995783 A 8/2017

(Continued)

OTHER PUBLICATIONS

Microfluidic device for DNA amplification of single cancer cells isolated from whole blood. Online, published date Sep. 16, 2025. Retrieved on Feb. 26, 2025 from URL: <https://www.youtube.com/watch?v=iLZihMi24n8>.*

(Continued)

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(57) **CLAIM**

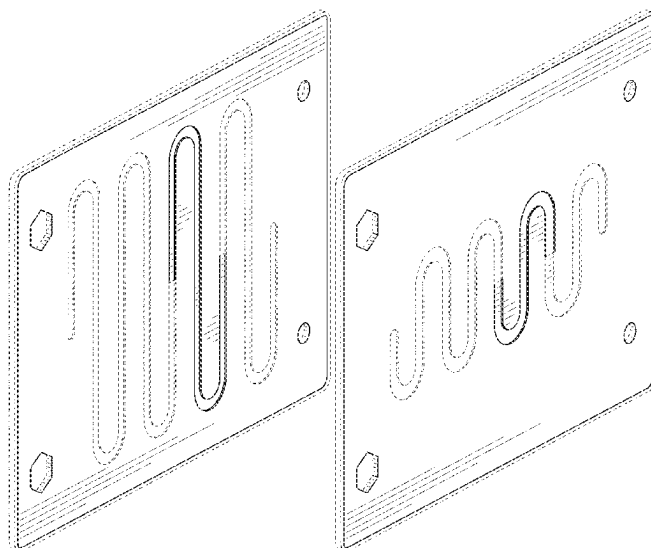
The ornamental design for a biological cell processing chip, as shown and described.

DESCRIPTION

FIG. 1 is an isometric view of a first embodiment of a biological cell processing chip showing our new design; FIG. 2 is a front elevation view thereof; FIG. 3 is a rear elevation view thereof; FIG. 4 is a right side elevation view thereof; FIG. 5 is a left side elevation view thereof; FIG. 6 is a top plan view thereof; FIG. 7 is a bottom plan view thereof; FIG. 8 is an isometric view of the first embodiment in a state of use; FIG. 9 is an isometric view of a second embodiment of a biological cell processing chip showing our new design; FIG. 10 is a front elevation view thereof; and, FIG. 11 is a rear elevation view thereof.

The dotted lines in the drawings depict boundaries of the claimed design that form no part thereof. The short-dashed lines in FIGS. 1-7 and 9-11 depict portions of the biological cell processing chip that form no part of the claimed design. The remaining short-dashed lines shown in FIG. 8 depict environmental subject matter and form no part of the claimed design.

1 Claim, 8 Drawing Sheets



Related U.S. Application Data

continuation-in-part of application No. 16/923,606, filed on Jul. 8, 2020, now Pat. No. 12,188,042.

(58) Field of Classification Search

CPC A61B 5/0002; A61B 5/042; A61B 5/0404;
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B01L 3/502715

See application file for complete search history.

(56) References Cited**U.S. PATENT DOCUMENTS**

6,074,605	A	6/2000	Meserol et al.	
6,481,648	B1 *	11/2002	Zimmermann ..	G01N 27/44717 239/690
6,602,791	B2 *	8/2003	Ouellet	B01L 3/502707 438/720
6,969,604	B1	11/2005	Yakovenko	
7,018,520	B2 *	3/2006	Voss	G01N 27/44743 204/453
D530,021	S *	10/2006	Muraca	D24/226
D621,060	S	8/2010	Handique	
D669,191	S	10/2012	Handique	
D879,999	S	3/2020	Wronko	
11,046,976	B2	6/2021	Wu	
D983,405	S *	4/2023	Huang	D24/224
11,654,431	B2 *	5/2023	Shartle	B01L 3/502715 422/400
D1,016,324	S *	2/2024	Chang	D24/225
2004/0197883	A1	10/2004	Dzekunov et al.	
2005/0118705	A1	6/2005	Rabbitt et al.	
2006/0057245	A1	3/2006	Haupt et al.	
2009/0166201	A1	7/2009	Tian et al.	
2010/0216126	A1	8/2010	Balachandran et al.	
2013/0256132	A1	10/2013	Chen	
2014/0332377	A1	11/2014	Su	
2017/0010259	A1	1/2017	Amoabediny et al.	
2017/0037357	A1	2/2017	Cattaruzzi et al.	
2017/0240945	A1	8/2017	Marquant et al.	
2017/0283761	A1	10/2017	Corso	
2019/0136224	A1	5/2019	Garcia Dominguez et al.	
2020/0017847	A1	1/2020	Chang	
2020/0221987	A1	7/2020	Liu	
2020/0238282	A1	7/2020	Burns et al.	
2020/0249232	A1	8/2020	Konstantopoulos et al.	
2020/0340014	A1	10/2020	Chang	
2022/0025402	A1	1/2022	Chang et al.	
2022/0088604	A1	3/2022	Yu et al.	

FOREIGN PATENT DOCUMENTS

CN	208250332	U	12/2018
JP	D2019-7688		2/2020
WO	WO-2006112870	A1	10/2006
WO	WO-2020014264	A1	1/2020
WO	WO-2021007315	A1	1/2021

OTHER PUBLICATIONS

Microfluidics Should Scare You. Online, published date Jun. 2019. Retrieved on Feb. 26, 2025 from URL: <https://www.usni.org/magazines/proceedings/2019/june/microfluidics-should-scare-you>.*

Introduction to Materials for Microfluidic Chips Fabrication. Online, published date 2017. Retrieved on Feb. 26, 2025 from URL: <https://www.elveflow.com/microfluidic-reviews/general-microfluidics/materials-for-microfluidic-chips-fabrication-a-review-2017/>.*

Andresen et al., "Injection molded chips with integrated conducting polymer electrodes for electroporation of cells," Journal of Micromechanics and Microengineering, 20: 1-9 (2010).

Boissel, L., et al., "Transfection with mRNA for CD19 specific chimeric antigen receptor restores NK cell mediated killing of CLL cells," Leuk Res., 33(9): 1255-1259 (2009).

Choi et al., "A high throughput microelectroporation device to introduce a chimeric antigen receptor to redirect the specificity of human T cells," Biomed Microdevices, 12: 855-863 (2010).

Ding et al., "High-throughput nuclear delivery and rapid expression of DNA via mechanical and electrical cell-membrane disruption," Nature biomedical engineering, 1(39): 1-7 (2017).

Examination Report from corresponding Australian Application No. 2020311908 dated Feb. 10, 2022.

International Preliminary Report on Patentability from PCT Application No. PCT/US2019/041055 dated Sep. 15, 2020.

International Search Report from corresponding PCT Application No. PCT/US2020/041188 dated Oct. 21, 2020.

International Search Report from PCT Application No. PCT/US2019/041055 dated Sep. 23, 2019.

Jiang, W., et al., "Functional characterization of interleukin-15 gene transduction into the human natural killer cell line NKL," Cytotherapy, 10(3): 265-274 (2008).

Li, S., "Electroporation Protocols: Preclinical and clinical gene medicine," Shulin Li (Ed.), Methods in Molecular Biology, 423: 1-511 (2008).

MaxCyte, "ExPERT GTx," (2020).

MaxCyte, "ExPERT STx," (2020).

MaxCyte, "MaxCyte VLX® Large-Scale Transfection System," (2017).

Moffett, H.F., et al., "Hit-and-run programming of therapeutic cytoreagents using mRNA nanocarriers," Nature Communications, 8(389): 1-13 (2017).

Office Action from corresponding Japanese Application No. 2021-501021 dated Apr. 5, 2022.

Office Action from corresponding U.S. Appl. No. 16/506,190 dated Sep. 16, 2021.

Selmeczi et al., "Efficient large volume electroporation of dendritic cells through micrometer scale manipulation of flow in a disposable polymer chip," Biomed Microdevices, 13: 383-392 (2011).

Selmeczi et al., "Large scale mRNA Transfection of Dendritic Cells by Electroporation in Ccontinuous Flow Systems," Methods in Molecular Biology, 1428 (Chapter 10): 151-161 (2016).

Wei, Z., et al., "A Laminar Flow Electroporation System for Efficient DNA and siRNA Delivery," Analytical Chemistry, 83: 5881-5887 (2011).

Young, J.L. & Dean, D.A., "Electroporation-Mediated Gene Delivery," Adv. Genet., 89: 49-88 (2015).

Zhao, D., et al., "A Flow-Through Cell Electroporation Device for Rapidly and Efficiently Transfecting Massive Amounts of Cells in vitro and ex vivo," Scientific Reports, 6: 1-9 (2016).

Book Chapter on Design and Fabrication of Microfluidic Devices Published. Online, published date Aug. 26, 2021. Retrieved on May 20, 2023 from URL: <https://people.sutd.edu.sg/~hashimoto/book-chapter-on-design-and-fabrication-of-microfluidic-devices-published/>.

* cited by examiner

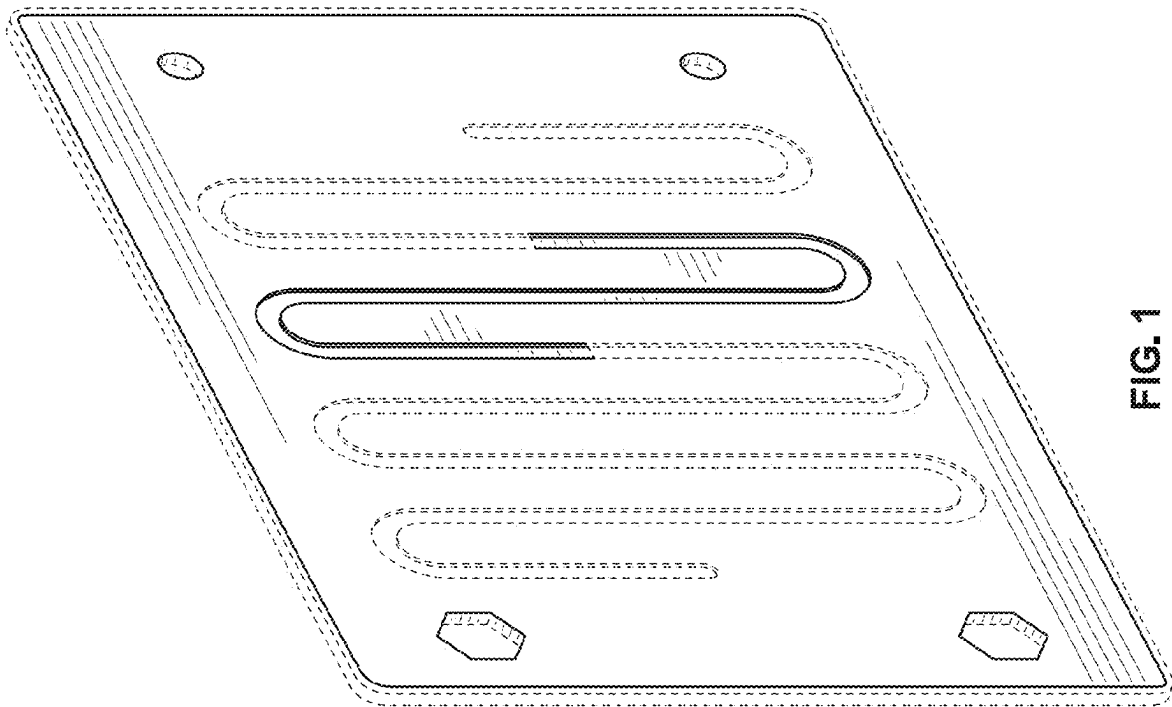
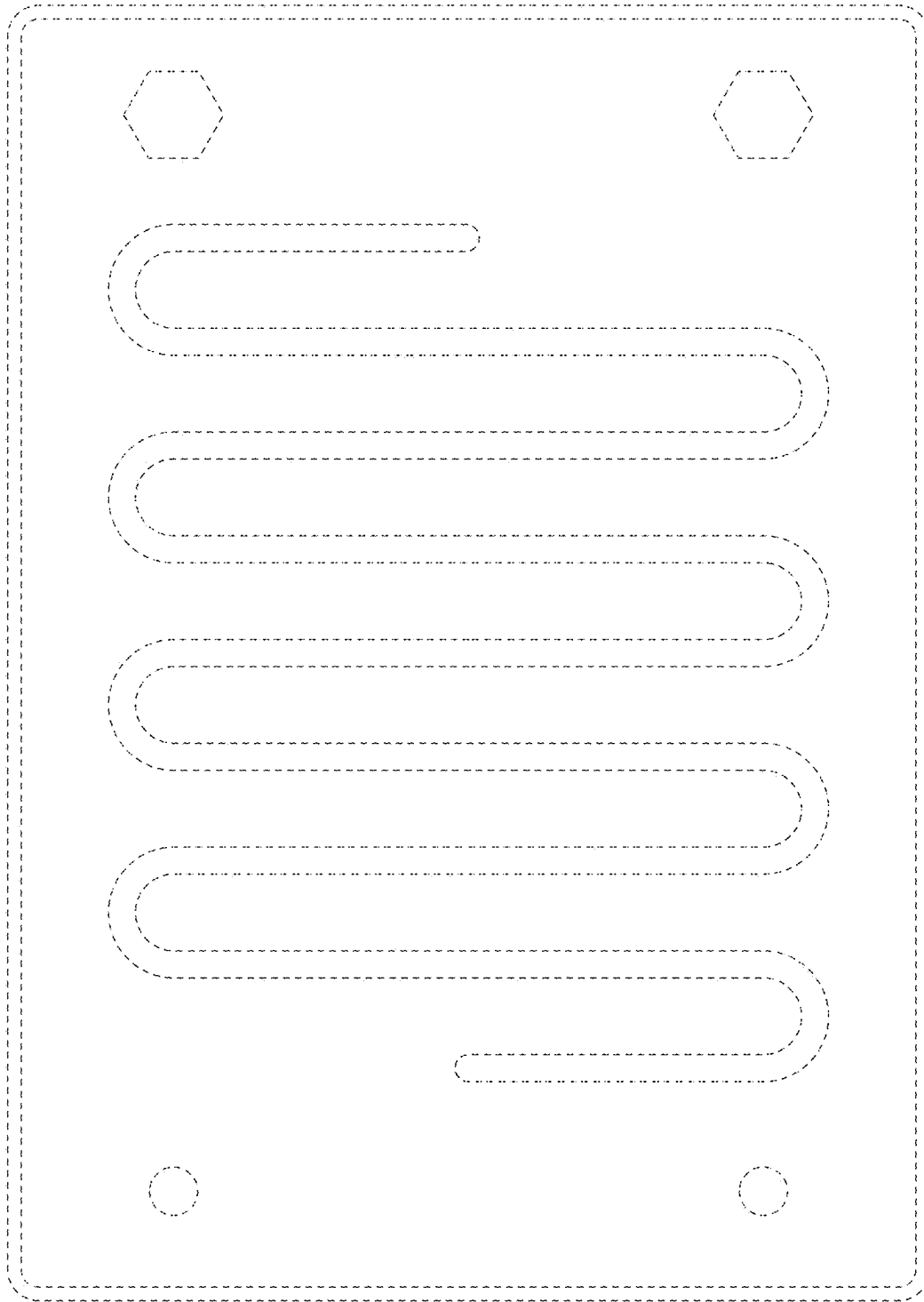


FIG. 1



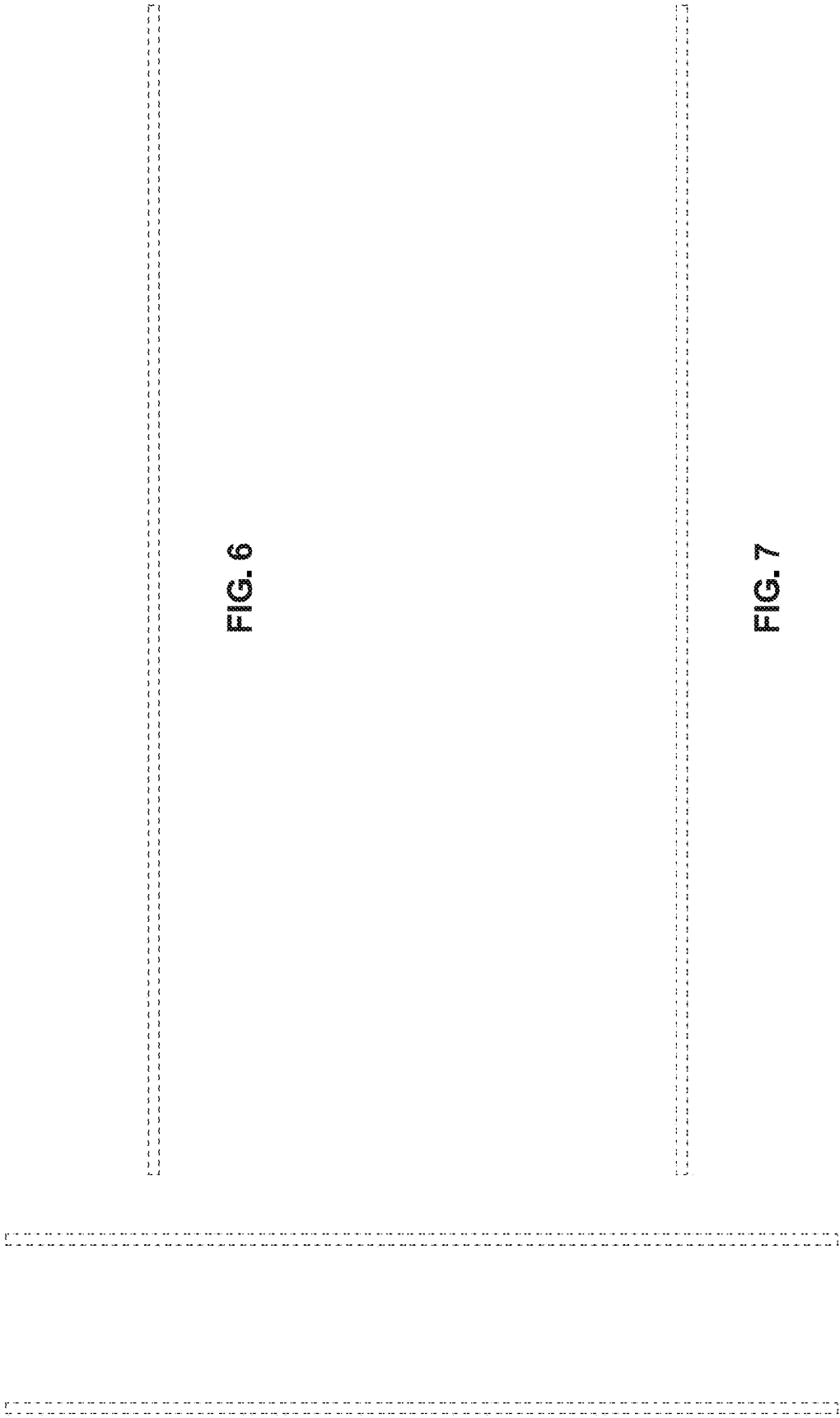
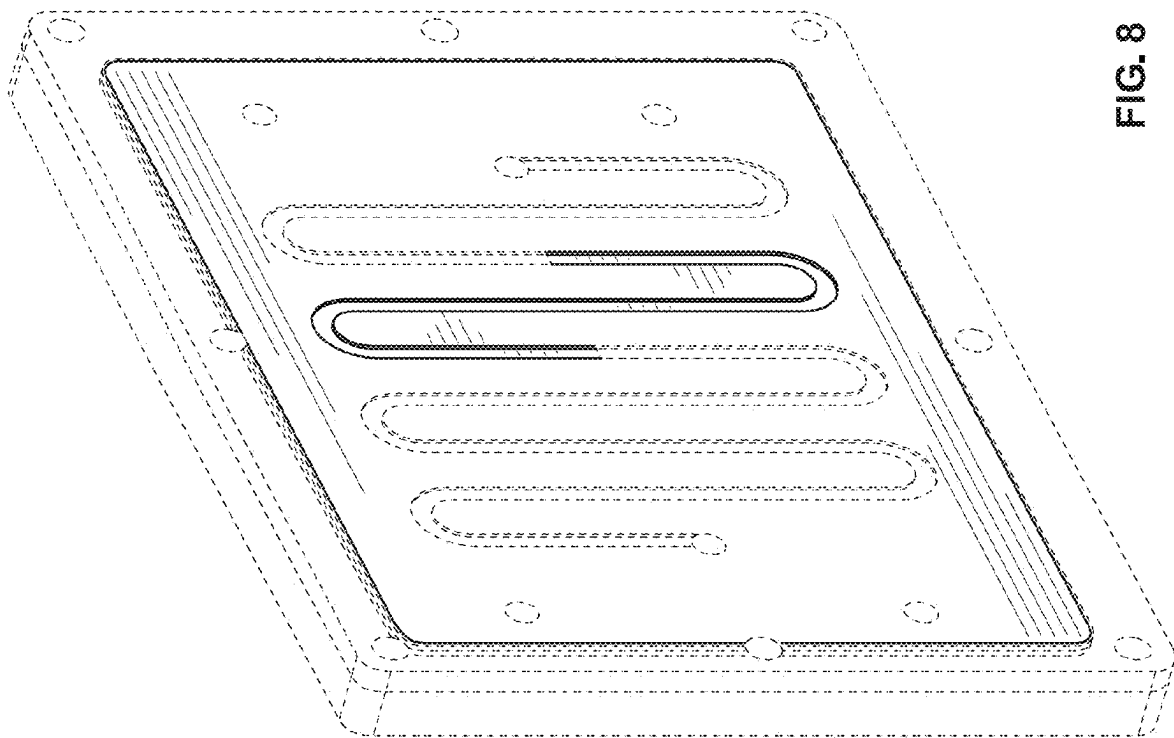
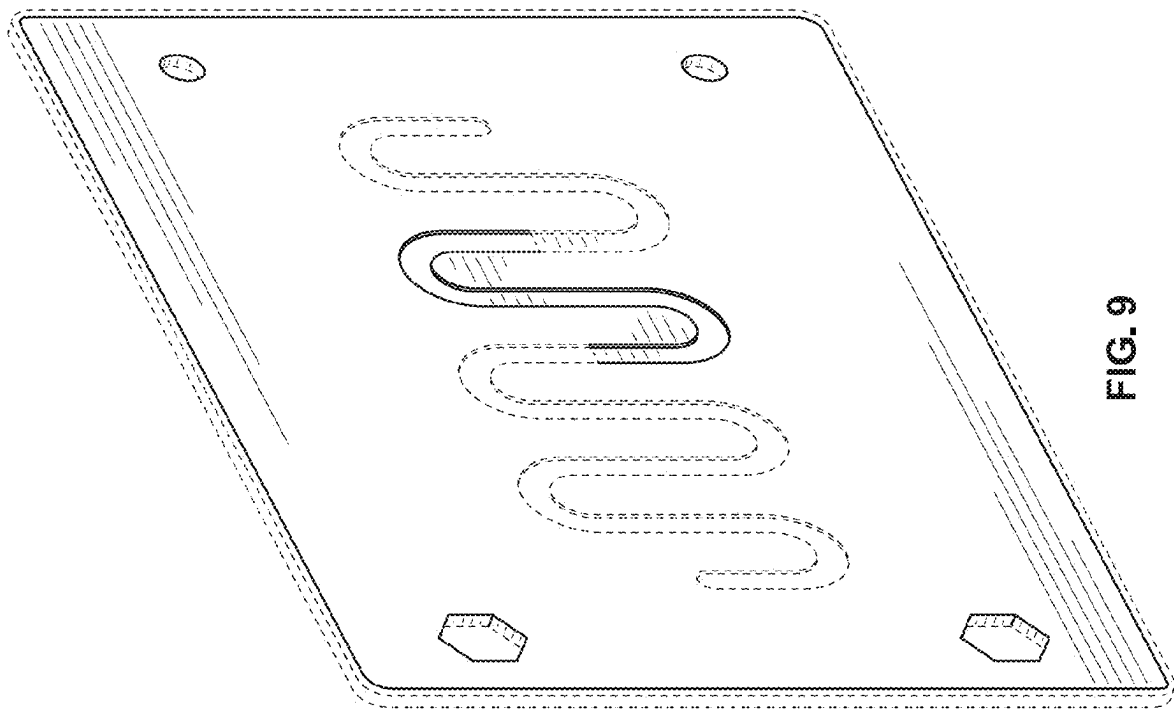


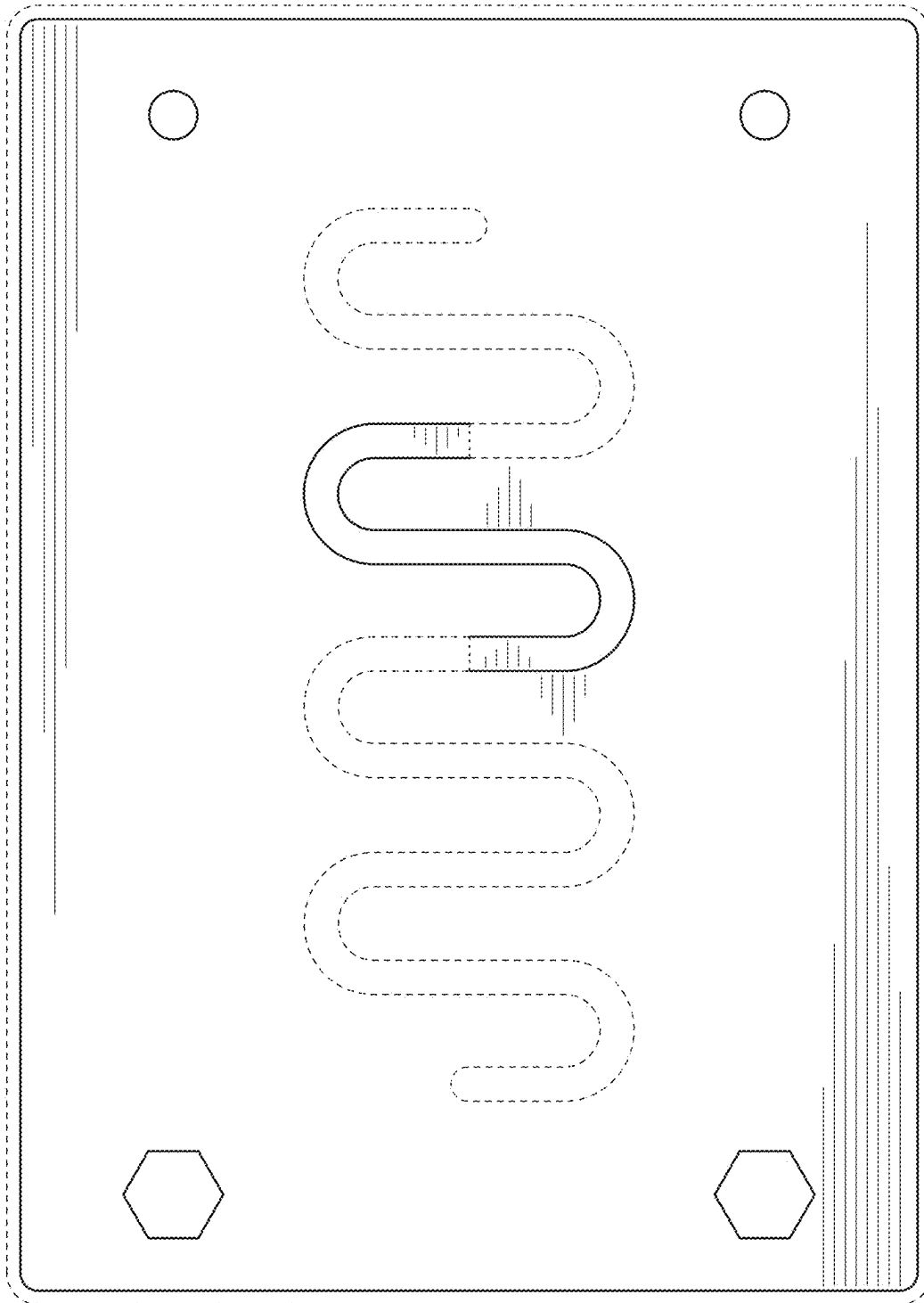
FIG. 6

FIG. 7

FIG. 4 FIG. 5







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